Alternating Direction Method of Multipliers

a talk by

Vinícius and Prof. Daniel Palomar
The Hong Kong University of Science and Technology

ELEC5470/IEDA6100A - Convex Optimization

December, 2020

Contents

1. Introduction
   Optimization algorithms, motivation
2. Alternating Direction Method of Multipliers
   The basics
3. Practical Examples
   Robust PCA and Graphical Lasso

Why use optimization algorithms?

Motivations

- large-scale optimization
  - machine learning/statistics with huge datasets
  - computer vision
- descentralized optimization
  - entities/agents/threads coordinate to solve a large problem by passing small messages

Optimization Algorithms

- Gradient Descent
- Newton
- Interior Point Methods (IPM)
- Block Coordinate Descent (BCD)
- Majorization-Minimization (MM)
- Block Majorization-Minimization (BMM)
- Successive Convex Approximation (SCA)

Optimization Algorithms

- Gradient Descent
- Newton
- Interior Point Methods (IPM)
- Block Coordinate Descent (BCD)
- Majorization-Minimization (MM)
- Block Majorization-Minimization (BMM)
- Successive Convex Approximation (SCA)
- ...

Optimization Algorithms

- Gradient Descent
- Newton
- Interior Point Methods (IPM)
- Block Coordinate Descent (BCD)
- Majorization-Minimization (MM)
- Block Majorization-Minimization (BMM)
- Successive Convex Approximation (SCA)
- ...
- **Alternating Direction Method of Multipliers (ADMM)**

Reference

- Boyd *et al.* **Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers**. *Foundations and Trends in Machine Learning*. 2010.
- available online for **free**: `https://web.stanford.edu/~boyd/papers/pdf/admm_distr_stats.pdf`
- citations: 13519[1]
- Yuxin Chen's Princeton lecture notes ELE 522: Large-Scale Optimization for Data Science

[1] as of Nov. 24th 2020

Dual Problem

- convex equality constrained optimization problem

$$\begin{array}{ll} \underset{\boldsymbol{x}}{\text{minimize}} & f(\boldsymbol{x}) \\ \text{subject to} & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \end{array}$$

- Lagrangian: $L(\boldsymbol{x}, \boldsymbol{y}) = f(\boldsymbol{x}) + \boldsymbol{y}^\top(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})$
- dual function: $g(\boldsymbol{y}) = \inf_{\boldsymbol{x}} \; L(\boldsymbol{x}, \boldsymbol{y})$
- dual problem: $\underset{\boldsymbol{y}}{\text{maximize}} \; g(\boldsymbol{y})$
- recover: $\boldsymbol{x}^\star = \underset{\boldsymbol{x}}{\operatorname{argmin}} \; L(\boldsymbol{x}, \boldsymbol{y}^\star)$

Dual Ascent

Dual Ascent

- gradient method for dual problem: $\boldsymbol{y}^{k+1} = \boldsymbol{y}^k + \rho^k \nabla g(\boldsymbol{y}^k)$
- $\nabla g(\boldsymbol{y}^k) = \boldsymbol{A}\boldsymbol{x}^{k+1} - \boldsymbol{b}$, where $\boldsymbol{x}^{k+1} = \arg\min_{\boldsymbol{x}} \; L(\boldsymbol{x}, \boldsymbol{y}^k)$
- dual ascent method is

$$\begin{aligned} \boldsymbol{x}^{k+1} &:= \arg\min_{\boldsymbol{x}} \; L(\boldsymbol{x}, \boldsymbol{y}^k) \\ \boldsymbol{y}^{k+1} &:= \boldsymbol{y}^k + \rho^k \left(\boldsymbol{A}\boldsymbol{x}^{k+1} - \boldsymbol{b}\right) \end{aligned}$$

Dual Ascent

- gradient method for dual problem: $\boldsymbol{y}^{k+1} = \boldsymbol{y}^k + \rho^k \nabla g(\boldsymbol{y}^k)$
- $\nabla g(\boldsymbol{y}^k) = \boldsymbol{A}\boldsymbol{x}^{k+1} - \boldsymbol{b}$, where $\boldsymbol{x}^{k+1} = \arg\min_{\boldsymbol{x}} \; L(\boldsymbol{x}, \boldsymbol{y}^k)$
- dual ascent method is

$$\begin{aligned} \boldsymbol{x}^{k+1} &:= \arg\min_{\boldsymbol{x}} \; L(\boldsymbol{x}, \boldsymbol{y}^k) \\ \boldsymbol{y}^{k+1} &:= \boldsymbol{y}^k + \rho^k \left(\boldsymbol{A}\boldsymbol{x}^{k+1} - \boldsymbol{b}\right) \end{aligned}$$

- why?

Dual Decomposition

Dual Decomposition

- suppose f is separable:

$$f(\boldsymbol{x}) = f_1(x_1) + \cdots + f_n(x_n), \boldsymbol{x} = (x_1, \ldots, x_n)$$

- then the Lagrangian is separable in $\boldsymbol{x}$:

$$L_i(x_i, \boldsymbol{y}) = f_i(x_i) + \boldsymbol{y}^\top \boldsymbol{a}_{*,i} x_i$$

- $\boldsymbol{x}$-minimization splits into n separate minimizations

$$x_i^{k+1} := \arg\min_{x_i} L_i(x_i, \boldsymbol{y}^k), i = 1, \ldots, n$$

which can be done in parallel and $\boldsymbol{y}^{k+1} = \boldsymbol{y}^k + \alpha^k \left(\sum_{i=1}^n \boldsymbol{a}_{*,i} x_i^{k+1} - \boldsymbol{b}\right)$

Optimization Problem

$$\begin{aligned} \underset{\boldsymbol{x},\ \boldsymbol{z}}{\text{minimize}} \quad & f(\boldsymbol{x}) + g(\boldsymbol{z}) \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} = \boldsymbol{c} \end{aligned} \tag{1}$$

- variables: $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{z} \in \mathbb{R}^m$
- parameters: $\boldsymbol{A} \in \mathbb{R}^{p \times n}$, $\boldsymbol{B} \in \mathbb{R}^{p \times m}$, and $\boldsymbol{c} \in \mathbb{R}^p$
- optimal value: $p^\star = \inf_{\boldsymbol{x},\boldsymbol{z}} \{f(\boldsymbol{x}) + g(\boldsymbol{z}) : \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} = \boldsymbol{c}\}$

Augmented Lagrangian Method

Augmented Lagrangian Method

- Augmented Lagrangian:

$$L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{y}) = \underbrace{f(\boldsymbol{x}) + g(\boldsymbol{z}) + \langle \boldsymbol{y}, \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} \rangle}_{\text{Lagrangian}} + \frac{\rho}{2} \|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_{\mathrm{F}}^2$$

- ALM consists of the iterations:

$$\boldsymbol{x}^{k+1}, \boldsymbol{z}^{k+1} := \underset{\boldsymbol{x},\boldsymbol{z}}{\arg\min}\ L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{y}^k) \text{ // primal update}$$

$$\boldsymbol{y}^{k+1} := \boldsymbol{y}^k + \rho \left(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z}^{k+1} - \boldsymbol{c}\right) \text{ // dual update}$$

- $\rho > 0$ is a penalty hyperparameter

Issues with Augmented Lagrangian Method

- the primal step is often expensive to solve – as expensive as solving the original problem
- minimization of $\boldsymbol{x}$ and $\boldsymbol{z}$ has to be done jointly

Alternating Direction Method of Multipliers

Alternating Direction Method of Multipliers

- Augmented Lagrangian:

$$L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{y}) = \underbrace{f(\boldsymbol{x}) + g(\boldsymbol{z}) + \langle \boldsymbol{y}, \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} \rangle}_{\text{Lagrangian}} + \frac{\rho}{2} \|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_{\mathrm{F}}^2$$

- ADMM consists of the iterations:

$$\begin{aligned}
\boldsymbol{x}^{k+1} &:= \arg\min_{\boldsymbol{x}} \; L_\rho(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{y}^k) \\
\boldsymbol{z}^{k+1} &:= \arg\min_{\boldsymbol{z}} \; L_\rho(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{y}^k) \\
\boldsymbol{y}^{k+1} &:= \boldsymbol{y}^k + \rho \left(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z}^{k+1} - \boldsymbol{c} \right)
\end{aligned}$$

- $\rho > 0$ is a penalty hyperparameter

Convergence and Stopping Criteria

- assume (very little!)
  - f, g are convex, closed, proper
  - L_0 has a saddle point
- then ADMM converges:
  - iterates approach feasibility: $\boldsymbol{A}\boldsymbol{x}^k + \boldsymbol{B}\boldsymbol{z}^k - \boldsymbol{c} \to \boldsymbol{0}$
  - objective approaches optimal value: $f(\boldsymbol{x}^k) + g(\boldsymbol{z}^k) \to p^\star$
- false (in general) statements: $\boldsymbol{x}$ converges, $\boldsymbol{z}$ converges
- true statement: $\boldsymbol{y}$ converges
- what matters: residual is small and near optimality in objective value

Convergence of ADMM in Practice

- ADMM is often slow to converge to high accuracy
- ADMM often converges to moderate accuracy within a few dozens of iterations, which is often sufficient for most practical purposes

Practical Examples

Robust PCA (Candes et al. '08)

- We would like to model a data matrix $\boldsymbol{M}$ as low-rank plus sparse components:

$$\begin{aligned} \underset{\boldsymbol{L},\, \boldsymbol{S}}{\text{minimize}} \quad & \|\boldsymbol{L}\|_* + \lambda \|\boldsymbol{S}\|_1 \\ \text{subject to} \quad & \boldsymbol{L} + \boldsymbol{S} = \boldsymbol{M} \end{aligned}$$

- where $\|\boldsymbol{L}\|_* := \sum_{i=1}^{n} \sigma_i(\boldsymbol{L})$ is the nuclear norm
- and $\|\boldsymbol{S}\|_1 := \sum_{i,j} |S_{ij}|$ is the entrywise ℓ_1-norm

Robust PCA via ADMM

ADMM for solving robust PCA:

$$
\begin{aligned}
\boldsymbol{L}^{k+1} &= \arg\min_{\boldsymbol{L}} \ \|\boldsymbol{L}\|_* + \operatorname{tr}\left(\boldsymbol{Y}^{k\top}\boldsymbol{L}\right) + \frac{\rho}{2}\left\|\boldsymbol{L} + \boldsymbol{S}^k - \boldsymbol{M}\right\|_{\mathrm{F}}^2 \\
\boldsymbol{S}^{k+1} &= \arg\min_{\boldsymbol{S}} \ \lambda\|\boldsymbol{S}\|_1 + \operatorname{tr}\left(\boldsymbol{Y}^{k\top}\boldsymbol{S}\right) + \frac{\rho}{2}\left\|\boldsymbol{L}^{k+1} + \boldsymbol{S} - \boldsymbol{M}\right\|_{\mathrm{F}}^2 \\
\boldsymbol{Y}^{k+1} &= \boldsymbol{Y}^k + \rho\left(\boldsymbol{L}^{k+1} + \boldsymbol{S}^{k+1} - \boldsymbol{M}\right)
\end{aligned}
$$

Robust PCA via ADMM

$$\boldsymbol{L}^{k+1} = \mathsf{SVT}_{\rho^{-1}}\left(\boldsymbol{M} - \boldsymbol{S}^k - \frac{1}{\rho}\boldsymbol{Y}^k\right)$$

$$\boldsymbol{S}^{k+1} = \mathsf{ST}_{\lambda\rho^{-1}}\left(\boldsymbol{M} - \boldsymbol{L}^{k+1} - \frac{1}{\rho}\boldsymbol{Y}^k\right)$$

$$\boldsymbol{Y}^{k+1} = \boldsymbol{Y}^k + \rho\left(\boldsymbol{L}^{k+1} + \boldsymbol{S}^{k+1} - \boldsymbol{M}\right),$$

where for any $\boldsymbol{X}$ with SVD $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, $\boldsymbol{\Sigma} = \mathsf{diag}\left(\{\sigma_i\}\right)$, we have

$$\mathsf{SVT}_\tau\left(\boldsymbol{X}\right) = \boldsymbol{U}\mathsf{diag}\left(\left\{(\sigma_i - \tau)^+\right\}\right)\boldsymbol{V}^\top$$

and

$$\left(\mathsf{ST}_\tau\left(\boldsymbol{X}\right)\right)_{ij} = \begin{cases} X_{ij} - \tau, & \text{if } X_{ij} > \tau, \\ 0, & \text{if } |X_{ij}| \leq \tau, \\ X_{ij} + \tau, & \text{if } X_{ij} < -\tau \end{cases}$$

Graphical Lasso

Precision matrix estimation from Gaussian samples:

$$\begin{aligned} \underset{\boldsymbol{\Theta}}{\text{minimize}} \quad & \underbrace{-\log\det\boldsymbol{\Theta} + \langle \boldsymbol{\Theta}, \boldsymbol{S} \rangle}_{\text{neg. log likelihood}} + \lambda \left\| \boldsymbol{\Theta} \right\|_1 \\ \text{subject to} \quad & \boldsymbol{\Theta} \succ \mathbf{0} \end{aligned}$$

Or equivalently, using a slack variable $\boldsymbol{\Psi} = \boldsymbol{\Theta}$

$$\begin{aligned} \underset{\boldsymbol{\Theta}, \boldsymbol{\Psi}}{\text{minimize}} \quad & \underbrace{-\log\det\boldsymbol{\Theta} + \langle \boldsymbol{\Theta}, \boldsymbol{S} \rangle}_{\text{neg. log likelihood}} + \lambda \left\| \boldsymbol{\Psi} \right\|_1 \\ \text{subject to} \quad & \boldsymbol{\Theta} \succ \mathbf{0}, \boldsymbol{\Theta} = \boldsymbol{\Psi} \end{aligned}$$

Graphical Lasso via ADMM

$$\boldsymbol{\Theta}^{k+1} = \underset{\boldsymbol{\Theta} \succ \mathbf{0}}{\arg\min} \; -\log\det\boldsymbol{\Theta} + \langle \boldsymbol{\Theta}, \boldsymbol{S} + \boldsymbol{Y}^k \rangle + \frac{\rho}{2} \left\| \boldsymbol{\Theta} - \boldsymbol{\Psi}^k \right\|_{\mathrm{F}}^2$$

$$\boldsymbol{\Psi}^{k+1} = \underset{\boldsymbol{\Psi}}{\arg\min} \; \lambda \left\| \boldsymbol{\Psi} \right\|_1 - \langle \boldsymbol{\Psi}, \boldsymbol{Y}^k \rangle + \frac{\rho}{2} \left\| \boldsymbol{\Theta}^k - \boldsymbol{\Psi} \right\|_{\mathrm{F}}^2$$

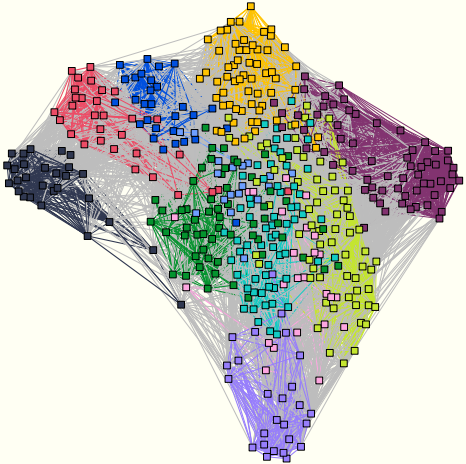
$$\boldsymbol{Y}^{k+1} = \boldsymbol{Y}^k + \rho \left(\boldsymbol{\Theta}^{k+1} - \boldsymbol{\Psi}^{k+1} \right)$$

Graphical Lasso via ADMM

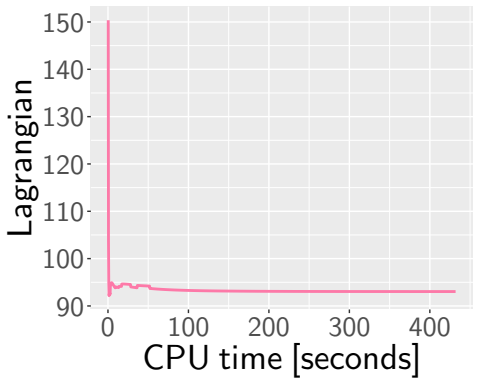
$$\boldsymbol{\Theta}^{k+1} = \mathcal{F}_\rho\left(\boldsymbol{\Psi}^k - \frac{1}{\rho}\left(\boldsymbol{Y}^k + \boldsymbol{S}\right)\right)$$
$$\boldsymbol{\Psi}^{k+1} = \mathsf{ST}_{\lambda\rho^{-1}}\left(\boldsymbol{\Theta}^{k+1} + \frac{1}{\rho}\boldsymbol{Y}^k\right)$$
$$\boldsymbol{Y}^{k+1} = \boldsymbol{Y}^k + \rho\left(\boldsymbol{\Theta}^{k+1} - \boldsymbol{\Psi}^{k+1}\right)$$

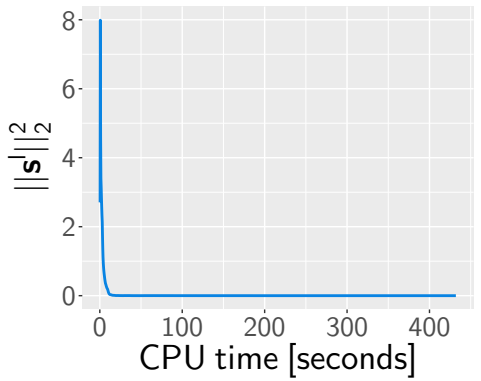
- where $\mathcal{F}_\rho(\boldsymbol{X}) := \frac{1}{2}\boldsymbol{U}\mathrm{diag}\left(\left\{\lambda_i + \sqrt{\lambda_i^2 + \frac{4}{\rho}}\right\}\right)\boldsymbol{U}^\top$, for $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$.

Network of stocks via Graphical Lasso



Network of stocks via Graphical Lasso





Conclusion

- ADMM is a versatile/flexible optimization framework
- may not be the best for a specific case, but often performs well in practice
- convergence often needs to be proved in a case-by-case scenario

Questions?